

PROJECT 3

APL Logistics Delivery Performance Analysis

Delivery Performance • Delay Risk • Logistics Efficiency

Project Area	Business Analytics / Supply Chain Analytics
Organization	Unified Mentor — Internship Project
Dataset Size	180,519 shipment records
Technology	Python • Pandas • NumPy • Plotly • Streamlit
Deployment	Interactive Streamlit Web Dashboard
Data Source	GitHub v1.0 Release

Project Objective

The project analyzes logistics delivery performance to identify late-delivery risk, shipping-mode differences, market and regional patterns, customer-segment risk, delivery-time variance, and operational areas requiring attention.

Business Problem

Large shipment datasets can hide operational bottlenecks when reviewed only through raw tables. The objective is to transform shipment-level data into an interactive decision-support dashboard that helps users compare risk and delivery behavior across operational dimensions.

Deliverable: A filterable Streamlit dashboard that converts 180,519 shipment records into executive KPIs, visual analysis, and data-driven action areas.

2. Data & Analytical Methodology

Dataset Overview

The dashboard uses the APL Logistics shipment dataset containing 180,519 records. The analysis uses shipment attributes such as shipping mode, order region, market, customer segment, actual shipping days, scheduled shipping days, late-delivery risk, and benefit per order.

Data Preparation

- Load the CSV dataset from the GitHub v1.0 release.
- Standardize the relevant fields and prepare analytical columns.
- Calculate delivery variance as actual shipping days minus scheduled shipping days.
- Use the late-delivery risk indicator to calculate shipment-level risk rates.
- Apply interactive filters before recalculating KPIs and visual summaries.

Core KPIs

KPI	Definition
Total Shipments	Number of records remaining after filters
At-Risk Shipments	Shipments flagged with late-delivery risk
Late-Risk Rate	$\text{At-risk shipments} \div \text{total filtered shipments} \times 100$
Avg Actual Days	Average actual shipping time
Avg Scheduled Days	Average scheduled shipping time
Avg Benefit / Order	Average benefit per shipment/order

Analytical Dimensions

The dashboard supports interactive analysis by Shipping Mode, Order Region, Market, and Customer Segment. This allows users to move from the overall portfolio view to focused operational slices without changing the underlying code.

3. Dashboard & Analysis

Executive KPI Overview

Metric	Full Dataset Result
Total Shipments	180,519
At-Risk Shipments	98,977
Late-Risk Rate	54.8%
Average Actual Shipping Days	3.50
Average Scheduled Shipping Days	2.93
Average Benefit / Order	21.97

Visual Analysis Modules

- Delivery Risk Distribution:** Shows the proportion of shipments classified as at-risk/late versus on-time/not at risk.
- Delivery Delay Profile:** Breaks shipments into Early, On Schedule, 1 Day Late, and 2+ Days Late categories.
- Shipping Mode Efficiency:** Compares late-delivery risk across First Class, Second Class, Same Day, and Standard Class.
- Market-Level Risk Analysis:** Compares late-risk rates across Europe, Pacific Asia, USCA, Africa, and LATAM.
- Regional Delivery Risk:** Highlights the highest-risk order regions using late-risk percentages.
- Customer Segment Analysis:** Compares late-delivery risk for Consumer, Corporate, and Home Office segments.
- Delivery Variance vs. Benefit:** Examines average benefit by the difference between actual and scheduled shipping days.

Interactive behavior was validated by selecting a single shipping mode. For example, First Class returned 27,814 shipments and a 95.3% late-risk rate, with all related KPIs and insights recalculating accordingly.

4. Key Findings & Business Insights

Overall Performance

Across the full dataset, 54.8% of shipments are flagged as late-delivery risk. There are 98,977 at-risk shipments out of 180,519 total records. Average actual shipping time is 3.50 days compared with 2.93 scheduled days.

Shipping Mode

The dashboard shows substantial variation by shipping mode. First Class records a 95.3% late-risk rate in the full selection, while Second Class, Same Day, and Standard Class show lower observed rates of 76.6%, 45.7%, and 38.1%, respectively.

Market & Regional Risk

At the market level, Europe has the highest observed late-risk rate among the displayed markets at 55.2%. At the regional level, Central Africa shows the highest observed late-risk rate at 58.0% among the selected regions.

Customer Segments

Late-risk rates are closely grouped across the three customer segments: Consumer 54.8%, Corporate 54.7%, and Home Office 55.1%. This indicates that delay risk is not confined to one customer segment in the full dataset.

Delivery Variance & Benefit

The variance analysis compares average benefit per order against the difference between actual and scheduled shipping time. The dashboard can therefore be used to investigate whether operational delay patterns coincide with differences in order-level benefit.

Validated Dynamic Insight

When only First Class is selected, the dashboard recalculates to 27,814 shipments, 26,513 at-risk shipments, 95.3% late-risk, 2.00 average actual days, 1.00 average scheduled days, and 23.12 average benefit per order. This confirms that the dashboard's calculations respond to user filters.

5. Recommendations, Conclusion & Deliverables

Data-Driven Action Areas

- **Shipping-mode monitoring:** Investigate operational drivers behind the high observed late-risk rate for First Class and review service allocation, capacity, and process adherence.
- **Market monitoring:** Use market-level risk comparisons to identify locations where routing, fulfillment, or carrier performance may require closer review.
- **Regional investigation:** Prioritize deeper root-cause analysis in high-risk regions such as Central Africa.
- **Schedule variance management:** Monitor cases where actual shipping time consistently exceeds scheduled time and investigate the operational bottlenecks behind the variance.
- **Customer experience:** Compare delay risk across customer segments before changing service levels, routing policies, or shipping-mode allocation.

Project Value

The project demonstrates how shipment-level operational data can be transformed into an interactive business analytics solution. Instead of relying on static reports, decision-makers can filter the dataset, compare risk dimensions, and observe KPI and insight changes in real time.

Technology Stack

Python: data processing and analytical logic | **Pandas:** data preparation | **NumPy:** numerical calculations | **Plotly:** interactive visualizations | **Streamlit:** dashboard deployment | **GitHub:** source control and dataset release

Conclusion

APL Logistics Delivery Performance Analysis provides a practical decision-support view of delivery risk, delay behavior, operational variation, and customer-facing implications. The final dashboard is interactive, filter-driven, and validated against the full 180,519-record dataset.

Submission Package

1. GitHub repository with source code and project files
2. GitHub v1.0 dataset release
3. Deployed Streamlit dashboard
4. Project technical/research documentation
5. Project feedback video

Prepared for Unified Mentor Business Analyst Internship — Project 3